

Felix Hoang

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Education

University of South Florida

Bachelor of Science in Computer Science, GPA: 3.7

Tampa, FL

Expected May 2028

Skills

Languages: Go, Python, SQL, Bash, YAML, C++

Cloud & Infrastructure: AWS, GCP, Kubernetes, Docker, Linux

Delivery & Automation: Terraform, Ansible, ArgoCD, Helm, GitHub Actions

Data & Backend: Kafka, gRPC, RabbitMQ, PostgreSQL, Redis, Apache Flink, MinIO, Debezium

Observability & Security: Prometheus, Grafana, Loki, Tempo, Alloy, Istio, HashiCorp Vault

Experience

Research Assistant

Sep 2025 – Present

Secure & Intelligent Systems Lab

Tampa, FL

- Engineered an automated data-preprocessing pipeline for Vision-Language Models (VLMs), utilizing **mcp-[arxiv-server](#)** to ingest and structure **500+** research papers into high-fidelity training corpora.
- Optimized the lab's Video Anomaly Detection (VAD) benchmarking suite by orchestrating an **n8n** and **GitHub Actions** CI/CD workflow, reducing model evaluation cycles from **4 hours to 2 minutes**.

Projects

Distributed Refurbished Marketplace | *Go, gRPC, PostgreSQL, RabbitMQ, Istio*

<https://github.com/phuchoang2603/refurbished-marketplace>

- Architected **Go** and **gRPC** microservices with a Database-per-Service pattern, preserving **100%** fault isolation and supporting **500+** simulated marketplace interactions per run.
- Streamlined **RabbitMQ** workflows for asynchronous state transitions, increasing storefront throughput by **40%** and cutting queue lag by **60%**.
- Operationalized **Istio** Service Mesh to enforce **mTLS** and circuit breaking, maintaining **99.9%** system availability during partial service degradation tests.
- Instrumented **LGTM** observability stack with distributed tracing, reducing Mean Time to Detection (MTTD) of cross-service bottlenecks by **50%** and surfacing incidents in under **1 minute**.

Fraud Detection MLOps Platform | *Python, Kafka, Flink, MinIO, ArgoCD*

<https://github.com/phuchoang2603/realtime-credit-card-fraud-detection>

- Built **Kafka** and **Flink** fraud inference pipeline handling **2,000+ QPS** with **P99 latency of 50ms** and **99.5%** event-processing accuracy.
- Implemented **Debezium** CDC to stream production database events into a **MinIO** data lake for offline retraining without impacting production load.
- Streamlined **ArgoCD** and **Helm** GitOps delivery to deploy in **15 minutes** with zero-downtime rolling updates.
- Enforced **Grafana** validation gates to reduce false-positive fraud flags by **15%** and alert noise by **30%**.

Zero-Trust RKE2 Infrastructure & Secrets Automation | *Terraform, Ansible, Vault, Proxmox, Tailscale*

<https://github.com/phuchoang2603/kubernetes-proxmox>

- Engineered **Terraform** and **Ansible** RKE2 automation on Proxmox VE, provisioning **12** specialized VMs and reducing manual setup time from **10 hours to 15 minutes**.
- Utilized ephemeral **GitHub Actions** runners to provision and bootstrap the homelab, authenticating to Proxmox with **JWT** and fetching secrets and configs from **HashiCorp Vault**.
- Configured **HashiCorp Vault** in **Terraform** as an **OIDC** provider, issuing short-lived **JWT** tokens that **kubelogin** uses to authenticate to the Kubernetes API server.
- Synchronized Kubernetes secrets with Vault KV engines using **External Secrets Operator (ESO)**, mitigating secret sprawl and keeping pod-level sensitive data rotation-capable.